- Short descriptions or updates from your sub-project
- Brief bios of team members, with headshots if available
- Photographs from fieldwork or other project activities
- Updates on workshops, meetings, presentations or other activities
- Any other stories, highlights or material that you think would work well on the website

Shorth description

Understanding Arctic Livestock Adaptation through genomic analyses

Our animal genomics team of Natural Resources Institute Finland (Luke) is investigating the biological basis of adaptation in livestock breeds that thrive in northern and Arctic environments. The current work focuses on genomic analyses of Northern Finncattle and other northern cattle breeds, such as Yakutian cattle, Eastern Finncattle and Western Finncattle, as well as reindeer and horse breeds adapted to harsh northern conditions. We combine whole-genome sequencing, transcriptomics (mRNA), microRNA profiling, and integrative bioinformatics approaches to identify genomic and regulatory mechanisms associated with environmental adaptation, resilience, metabolism, and thermoregulation.

The teams integrate genomic and transcriptomic datasets from native cattle populations, including Northern Finncattle, and compare them with breeds from diverse climatic regions. We are also developing multi-species approaches that enable comparative analyses across cattle, reindeer and horses, providing new insights into convergent mechanisms of adaptation to northern environments. In addition, we are participating in the newly established Global Consortium for the Conservation of Bovine Genetic Resources. Moreover, the team is preparing new scientific publications on whole-genome sequencing and miRNA regulation in gene expression of metacarpal adipose tissues of cattle, while exploring advanced multi-omics integration and machine learning approaches for future analyses.

Team members, The animal genomics team of Natural Resources Institute Finland (Luke)

Juha Kantanen, Professor of Livestock Genomics at the Natural Resources Institute Finland (Luke). Professor Kantanen has participated in numerous national and international research projects focusing on animal genetic resources, molecular and population genetics, animal genomics, and the valuation of animal genetic resources. He is a leading expert in multidisciplinary research on livestock genetic resources across northern Eurasia.



Kisun Pokharel, Senior Scientist in animal genomics and bioinformatics at the Natural Resources Institute Finland (Luke). His research focuses on genome assembly, functional genomics and population genomics in livestock, reindeer and fish. He applies advanced sequencing technologies and computational approaches to investigate genetic diversity, adaptation and biologically important traits, with particular emphasis on native northern breeds and species.



Rodney Okwasiimire, Doctoral Researcher at the University of Helsinki and the Natural Resources Institute Finland (Luke). Building on previous experience in molecular diagnostics and epidemiology of livestock diseases, his doctoral research integrates whole-genome sequencing data analysis, genome assembly, and population genomics to investigate the genetic diversity and genomic basis of adaptation in indigenous African cattle.



Melak Weldenegodguad, Research Scientist at the Natural Resources Institute Finland (Luke). His work combines genomics, transcriptomics, and bioinformatics to investigate adaptation, biological diversity, and complex traits in domesticated and wild animal populations. He has experience in the analysis of whole-genome sequencing, RNA-seq, miRNA-seq, and ancient DNA datasets, with research focusing on native livestock breeds and wild animal populations across northern Eurasia, including cattle, reindeer, horses, sheep, and chickens. His current interests include multi-omics integration, functional genomics, environmental adaptation, and the use of machine learning approaches to improve the understanding of biological mechanisms and livestock resilience in changing environments.



Jonne Pohjankukka, senior scientist and coordinator of artificial intelligence method services at the Natural Resources Institute Finland (Luke), graduated from University of Turku with a major in data science and machine learning in 2018. He has experience working in the research, development and management of computer vision and artificial intelligence related projects both in the public and private sectors. In recent years, he has worked in a variety of sensor fusion based applications involving remotely sensed big data with end goals of creating situational awareness and automated data processing systems. Examples of his work include for example Rolls-Royce ship intelligence and Luke Naturewatch projects.



Workshops, meetings, presentations or other activities

- Kisun Pokharel: Workshop “Machine Learning for Multi-Omics Integration”, 15–17 December 2025 at the University of Barcelona, Faculty of Biology (Dept. Evolutionary Biology, Ecology & Environmental Sciences, IRBio), Barcelona, Spain.
- Juha Kantanen: The International Symposium on Bovine Genetic Resources 2026 16-18 September 2026, Jilin Agricultural University, Changchun, PR of China. Oral presentation by Juha Kantanen: Northern Eurasian Bovine Genetic Resources.

ADDITIONAL PHOTOS

Northern Finncattle photos

The research focuses on Northern Finncattle and other locally adapted northern farm animal breeds and reindeer, emphasizing their genetic diversity and cultural and societal

significance. Northern Finncattle narrowly escaped extinction in the 1980s. The current population consists of approximately 850 breeding cows, with the principal conservation herd maintained at the educational farm of the Vocational College Lappia in Tervola, Finnish Lapland. In addition, embryos and semen from Northern Finncattle bulls are stored in the national animal genetic resources cryobank.



Photo: Annu Puurunen



Photo: Juha Kantanen



Photo: Luke Archive